

LC 174: Dungeon Game

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Companies: Darwinbox, Flipkart

Intuition: unlike lc 64, we don't care about finding minimum sum, we need to ensure knight never dies at any intermediate cell.

KEY CONCEPT

at every cell, find how much health is needed to enter that cell



need at 5 = 6 (start from end)

need at 4 = (6 - 1) = 5

need at 3 = (5 - 3) = 2

need at 2 = 2 - (-3) = 5

need at 1 = 5 - (-2) = 7

hence return the need at 1st cell

Edge Case Consideration:

now imagine if at 4 there was 60

then need at 4 = (6 - 60) = -ve

it means 60 was already enough, hence need minimum need, which is 1

C++ Implementation (BFS Approach)

```
class Solution {
public:
    int calculateMinimumHP(vector<vector<int>>& dungeon) {
        int n=dungeon.size();
        int m=dungeon[0].size();

        queue
        <
        int,int>> q;
        vector<vector<int>> seen(n,vector<int>(m,0));
        vector<vector<int>> minu(n,vector<int>(m,INT_MAX));
        //to track min hp required to enter a cell
        //we will go from aadya to my start pos

        minu[n-1][m-1]=max(1,1-dungeon[n-1][m-1]);
        //last pos pe atleast 1 health leke pohochna he
        //case 1 -> -ve then eg -69 then atleast 70 hona chahiye last cell 1-(-69)
        //case 2-> +ve then eg +67 then already enough 1-67=-66 , negative val
        leke cell me no entry
        //hence atleast 1

        q.push({n-1,m-1});
        seen[n-1][m-1]=1;

        //aadya princess se start krte krte start pos tak jaaunga

        vector
        <
        int,int>> dir={{-1,0},{0,-1}}; //up and left

        while(!q.empty()){
            auto it=q.front();
            q.pop();

            int r=it.first;
            int c=it.second;

            for(auto &d:dir){
                int nr=r+d.first;
                int nc=c+d.second;

                if(nr<0 || nc<0 || nr>=n || nc>=m) continue;

                int need=max(1,minu[r][c]-dungeon[nr][nc]);
                minu[nr][nc]=min(minu[nr][nc],need);
            }
        }
    }
};
```